Cloud Native Data Engineering

Week 2. Cloud Services and Architecture

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Lecture Outline

1. Cloud Networking
2. Identity and Security
3. Compute & Serverless Services
4. Storage, Messaging & Databases
5. Analytics, AI/ML & Marketplace Services
6. Cloud Architecture Principles
7. Reference Architectures
8. System Design

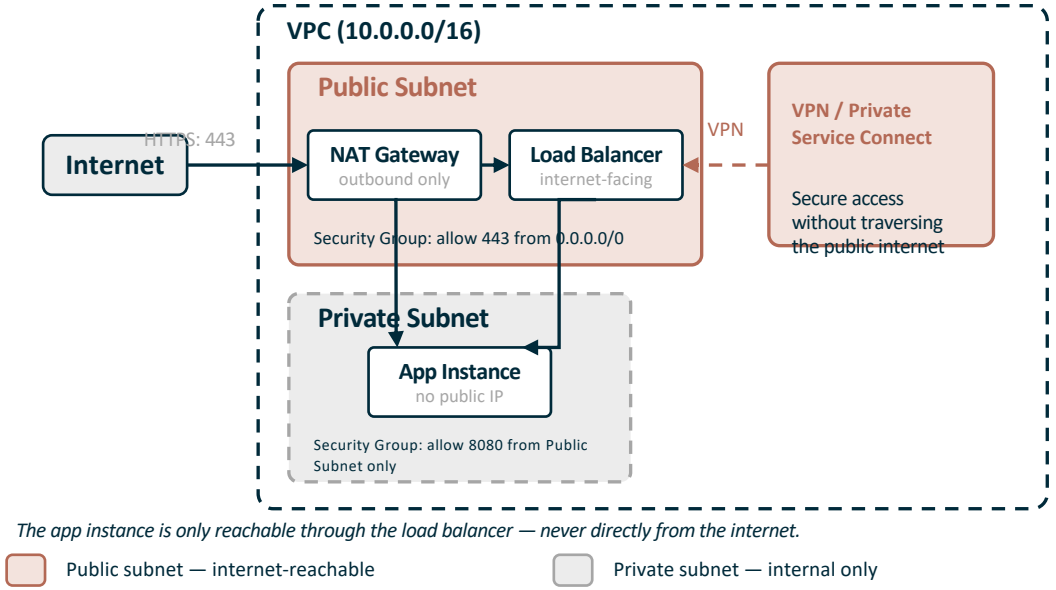


Cloud Networking

Module 1

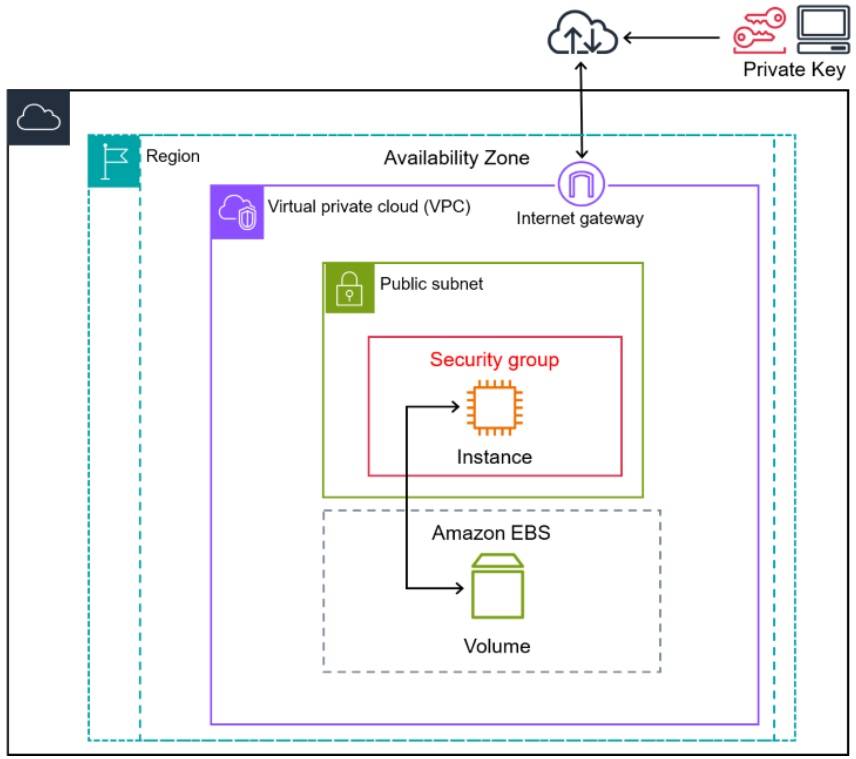
Cloud Networking

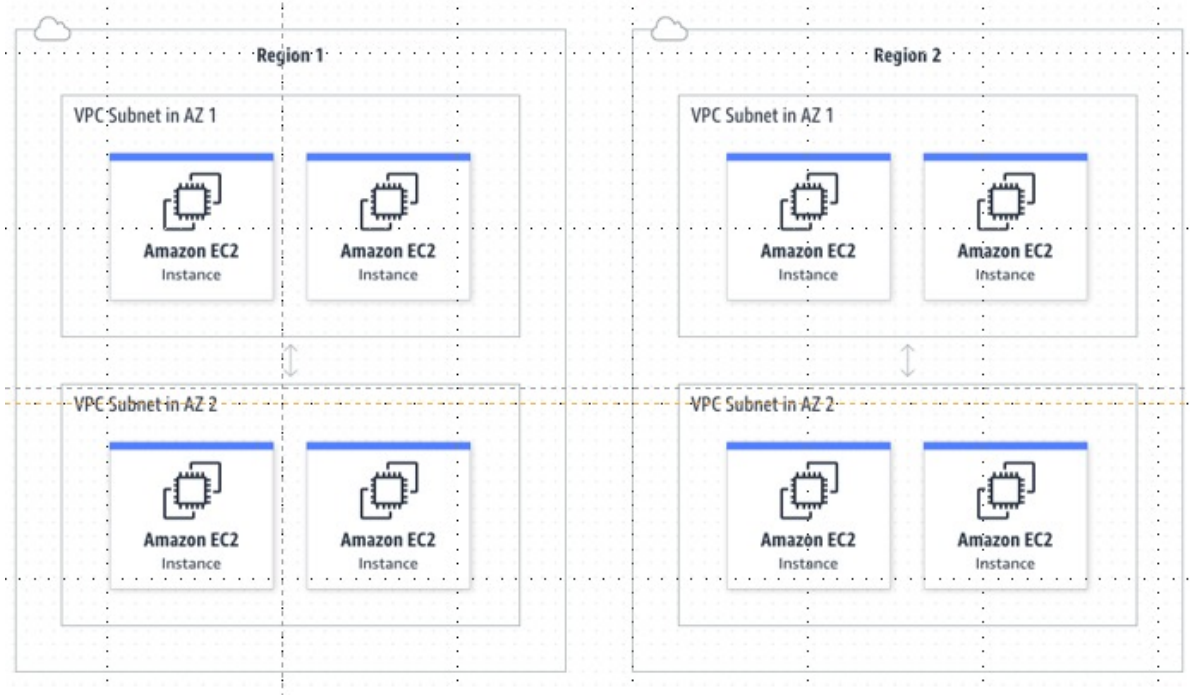
- VPC (Virtual Private Cloud)
  - Logically isolated network within the cloud
- Subnets
  - Subdivide VPC into public (internet-facing) and private (internal) zones
- Security Groups / Firewall Rules
  - Stateful packet filtering at instance level
- NAT Gateway
  - Allows private subnet instances to reach the internet for updates



Virtual Private Cloud (VPC)

- Foundational service that lets you launch cloud resources in a logically isolated virtual network that you define.

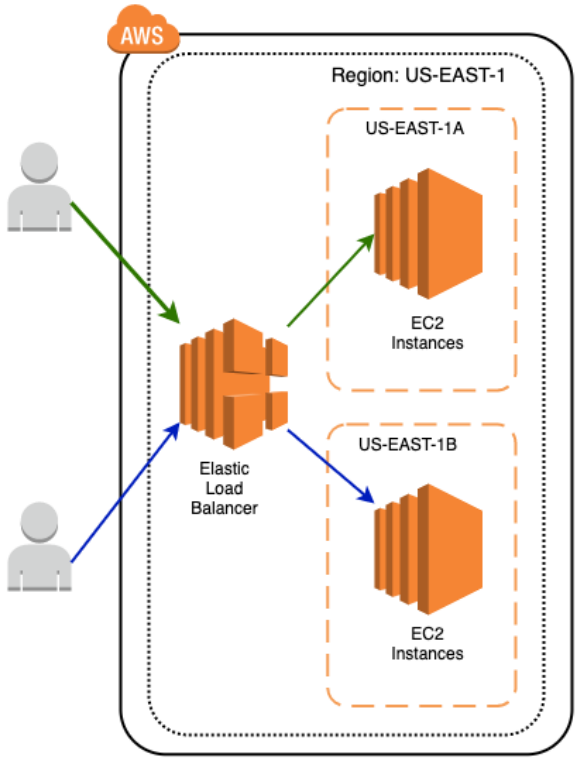




https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html

Load Balancing

- Distributing a set of tasks over a set of resources with the aim of making their overall processing more efficient
- Types
  - Application Load Balancer
  - Network Load Balancer
  - Gateway Load Blancer

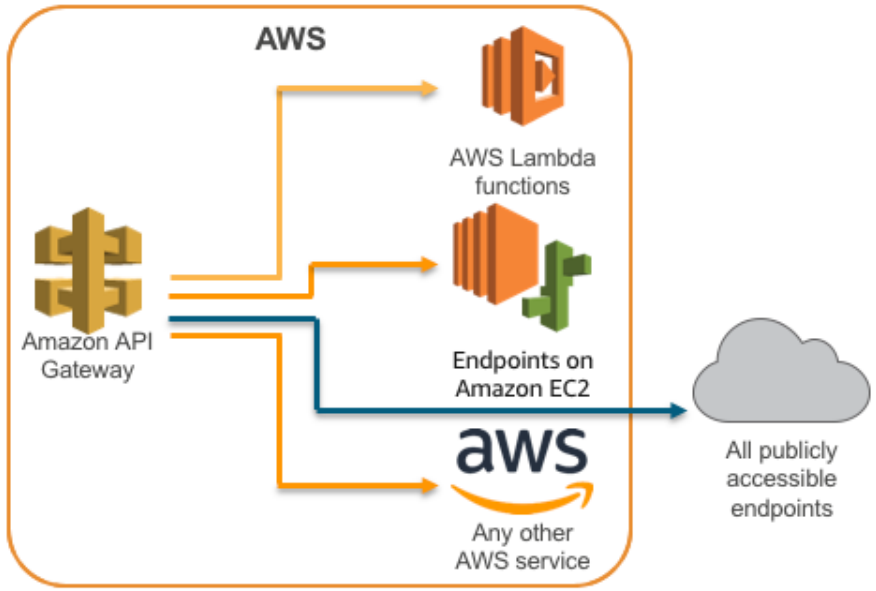


API Services

- Fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure Application Programming Interfaces (APIs) at any scale

Amazon API Gateway

API Gateway



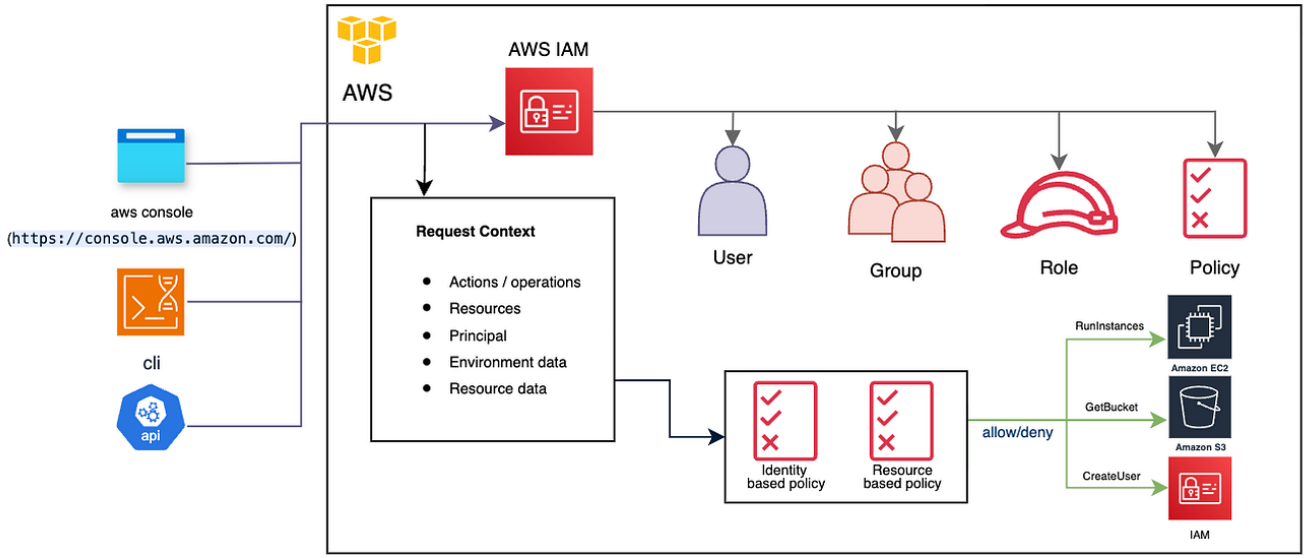


Identity and Security

Module 2

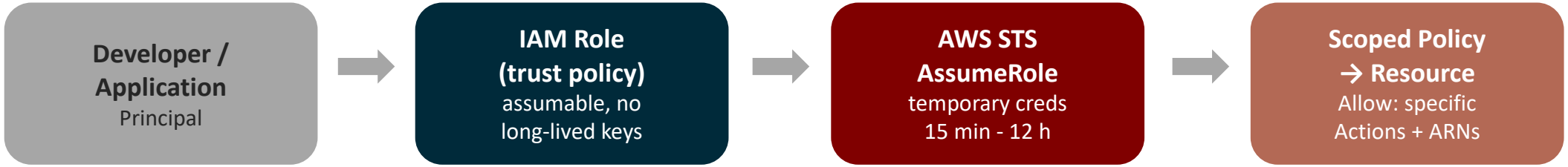
Identity & Access Management (IAM)

- Web service that helps you securely control access to cloud resources.
- With IAM, you can manage permissions that control which cloud resources users can access.



IAM: Concepts

- User: long-lived identity for a person or application (avoid for services)
- Group: collection of users that share a set of permissions
- Role: assumable identity with no long-lived credentials; preferred for services
- Policy: JSON document defining allowed/denied actions on resources



Least privilege: the role's policy grants only the specific Actions and Resources the workload needs — nothing more.

IAM Best Practices

- Never use the root account for daily work: root has unrestricted access; compromise is catastrophic
- Use roles for services, not IAM users: roles have no long-lived credentials to leak
- Apply least privilege from day one: easier to add permissions than to remove them after an incident
- Enable MFA for all human users: protects against credential theft
- Rotate access keys regularly: limits the window of exposure for any leaked key

Network & Data Security

- Public subnet: load balancers and NAT gateways only
- Private subnet: application/compute servers - no direct internet exposure
- Data subnet: RDS, ElastiCache - never publicly accessible
- Security groups are stateful firewalls: inbound denied by default; return traffic is auto-allowed
- Reference security-group IDs (not CIDRs) to express trust between tiers
- Use VPC endpoints for S3/DynamoDB to avoid NAT gateway charges



Compute & Serverless Services

Module 2

Compute Services

- Compute Infrastructure for creating Virtual Machines of various capacities
- General Purpose, Compute Optimized, Memory Optimized, HPC Optimized, etc.



https://aws.amazon.com/ec2/instance-types/



https://cloud.google.com/compute/docs/machine-resource

Virtual Compute

- Virtual Machines: flexible but requires OS management; use for custom workloads
- Machine families: general-purpose, compute-optimized, memory-optimized, GPU/TPU
- Auto-scaling groups: automatically add/remove instances based on load
- Preemptible / Spot instances: 60-90% cheaper; can be interrupted; good for batch jobs
- Right-sizing: match instance type to workload; over-provisioning wastes money

Serverless Compute

- Serverless event-driven compute service that lets users run code for most types of application or backend service without provisioning or managing servers







Serverless Compute

- Functions as a Service (FaaS): code runs on-demand; no server management
- AWS Lambda, GCP Cloud Functions, Azure Functions
- Ideal for event-driven ingestion, lightweight transformations, API backends
- Cold start latency: first invocation after idle period incurs startup overhead
- Execution limits (15 min AWS, 9 min GCP); not suitable for long-running jobs
  - Course lab: AWS Lambda function with SAM template (template.yaml); Streamlit UI as the frontend
  - boto3 SDK: Python library for all AWS services — S3, EC2, RDS, DynamoDB, Lambda



Storage, Messaging & Databases

Module 3

Storage Services

- Scalable, secure, and cost-effective solutions for storing and managing data in the cloud
- Data Redundancy and Disaster Recovery, Integration with Other Cloud Services, Data Lifecycle Management, Automatic Updates and Maintenance







Object Storage

- Flat namespace of objects (files) identified by keys (paths)
- AWS S3, GCP Cloud Storage, Azure Blob Storage
- Virtually unlimited capacity; durability 99.999999999% (11 nines)
- Storage classes: Standard (hot) → Nearline → Coldline / Glacier (archival)
- Use object storage as the foundation of the data lake and lakehouse

Pub Sub Services

- Cloud Pub/Sub is a fully-managed, scalable asynchronous messaging service that allows decoupled and distributed systems to exchange messages reliably
- Facilitates real-time communication and data streaming between independent systems and applications.

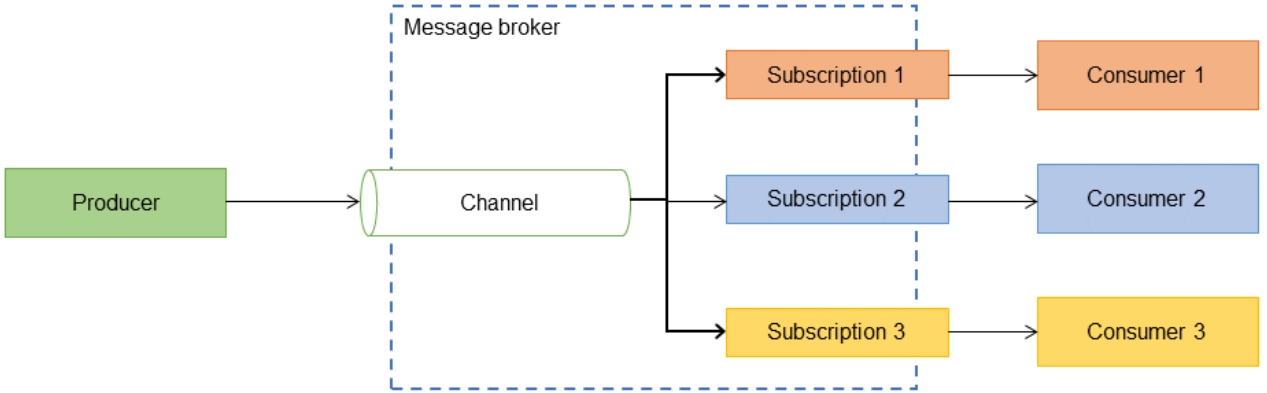
Amazon SNS

Amazon SQS

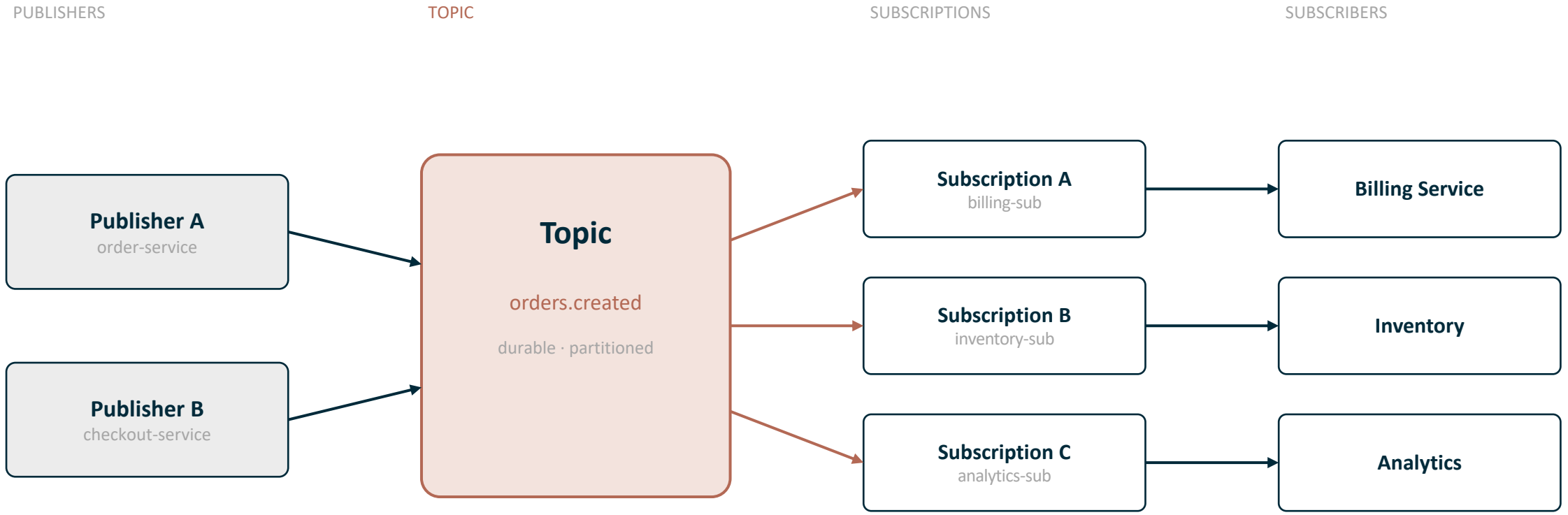
AWS EventBridge

amazon Kinesis

Google Cloud Pub/Sub



Messaging and Pub/Sub



Each subscriber gets its own independent copy — publishers and subscribers are fully decoupled

Cloud Databases

- Scalable and reliable storage solutions for structured and unstructured data while leveraging the flexibility and scalability of cloud computing infrastructure.



















Analytics, AI/ML & Marketplace Services

Module 4

Cloud Data Warehouse (DW)

- Fully-managed petabyte-scale data warehouse services that makes it easy to analyze data using standard SQL and existing Business Intelligence (BI) tools.



Business Intelligence (BI)

- BI tools enable you to create powerful visualizations and dashboards so that managers and leaders can make data-driven decisions.











Big Data

- Managed services for storing and processing huge amounts of data (big data)
- Support for open-source tools such as Apache Spark, Hadoop, and Hive.











Data Cataloging

- Centralized metadata management, data discovery, and governance capabilities to organize and understand data assets stored across various cloud sources.



AWS Glue
Data Catalog



Google



Machine Learning (ML)

- Managed ML services that make it easy to train, and deploy ML models at scale
- Support Model Governance, Pipelines, Deployment, Registry, Monitoring, etc.



Amazon SageMaker



vertex.ai



Azure Machine Learning

Natural Language Processing (NLP)

- Managed NLP services that support Speech recognition, Text translation, Text-to-speech, Chatbot, Text analysis and insights, etc.



Amazon Comprehend



Amazon Transcribe



Amazon Polly



Amazon Textract



Amazon Lex



Amazon Translate

Computer Vision (CV)

- Managed Services that support computer vision problems such as image recognition, object detection





Artificial Intelligence (AI) Services

- Managed AI services that enable conversational bots, agents, and LLMs

Foundational



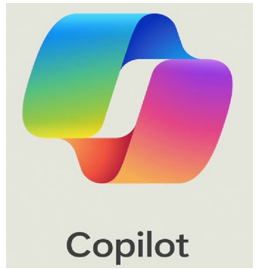




Conversational







Data Marketplaces

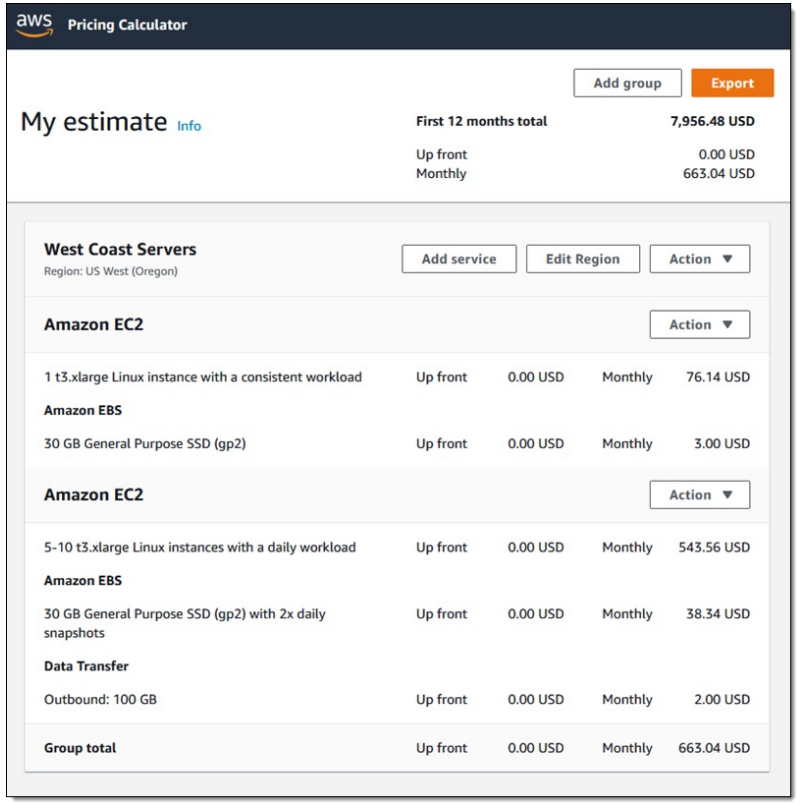
- Online location or store that facilitates the buying and selling of data between producers and consumers across organizational boundaries

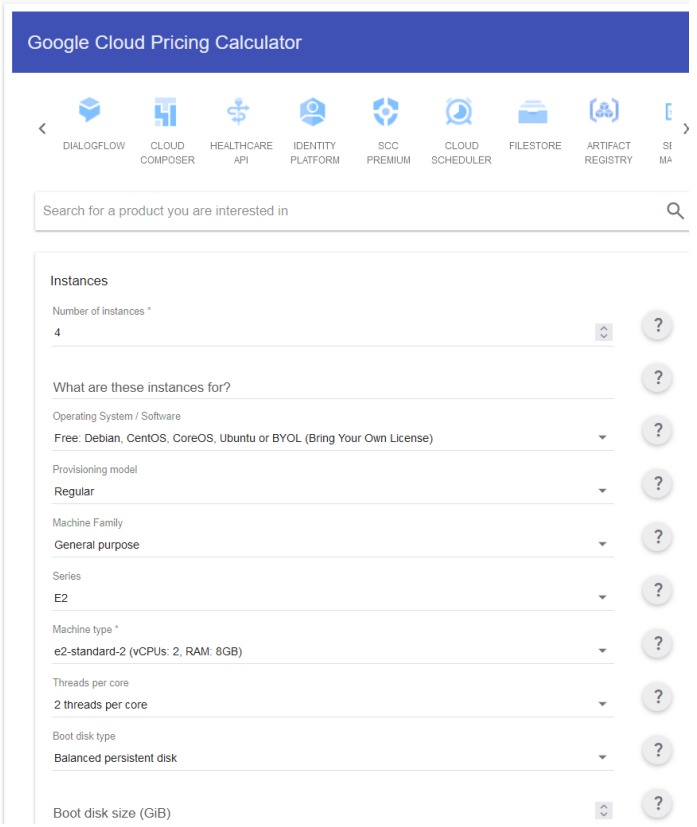


AWS Data Exchange

Analytics Hub

Pricing Calculators





Resource Cleanup Checklist

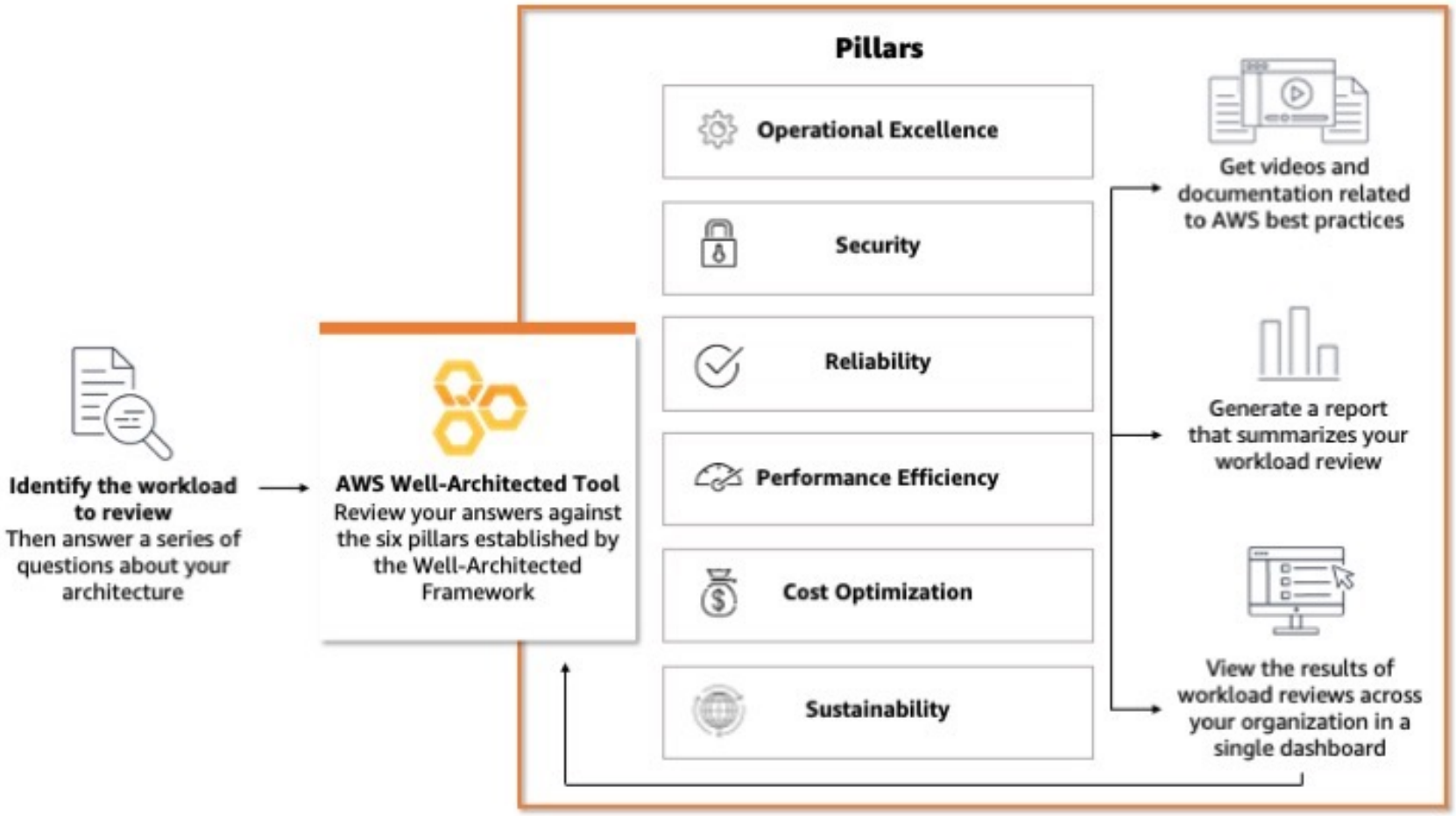
1. Terminate / stop compute — EC2 instances, test Lambda resources
2. Empty & delete S3 buckets created for the lab
3. Delete RDS instances & manual snapshots
4. Detach & delete IAM roles / policies created for the lab
5. Delete security groups & VPC extras (NAT gateways, endpoints)
6. Confirm Cost Explorer forecast returns to $0 for the lab tag



Cloud Architecture Principles

Module 6

Well Architected Framework (WAF)



https://aws.amazon.com/architecture/well-architected/

1. Operational Excellence

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operational failures

https://aws.amazon.com/architecture/well-architected/

2. Security

- Implement a strong identity foundation
- Enable traceability
- Apply security at all layers
- Automate security best practices
- Protect data in transit and at rest
- Keep people away from data
- Prepare for security events

https://aws.amazon.com/architecture/well-architected/

3. Reliability

- Automatically recover from failure
- Test recovery procedures
- Scale horizontally to increase aggregate workload availability
- Stop guessing capacity
- Manage change in automation

https://aws.amazon.com/architecture/well-architected/

4. Performance Efficiency

- Democratize advanced technologies
- Go global in minutes
- Use serverless architectures
- Experiment more often
- Consider mechanical sympathy

https://aws.amazon.com/architecture/well-architected/

5. Cost Optimization

- Implement Cloud Financial Management
- Adopt a consumption model
- Measure overall efficiency
- Stop spending money on undifferentiated heavy lifting
- Analyze and attribute expenditure

https://aws.amazon.com/architecture/well-architected/

Cost Management: Budgets, Tagging & Cost Explorer

- Cost Explorer: visualize and analyze historical spend by service, account, and tag
- Budgets: set alerts when spend exceeds a threshold (e.g. 80% of expected monthly cost)
- Tagging: allocate cost to projects/environments - tag every resource at creation
- Cleanup automation: delete idle resources; use TTL/ExpiresOn tags to enforce expiry
- Cost discipline matters from day one - almost every cloud resource is billed

General Design Principles

- Stop guessing your capacity needs
- Test systems at production scale
- Automate to make architectural experimentation easier
- Allow for evolutionary architectures
- Drive architectures using data
- Improve through game days

https://aws.amazon.com/architecture/well-architected/

Summary

- **Network:** Use VPCs, subnets, security groups, NAT, and load balancers to control traffic and isolation.
- **Identity:** Prefer roles and temporary credentials; enforce least privilege and MFA.
- **Compute:** Use VMs for control and serverless for event-driven workloads.
- **Data:** Build around object storage; use messaging to decouple systems.
- **Services:** Match warehouses, big-data, catalogs, and AI/ML tools to the access pattern.
- **Architecture:** Use the Well-Architected Framework while continuously managing cost.

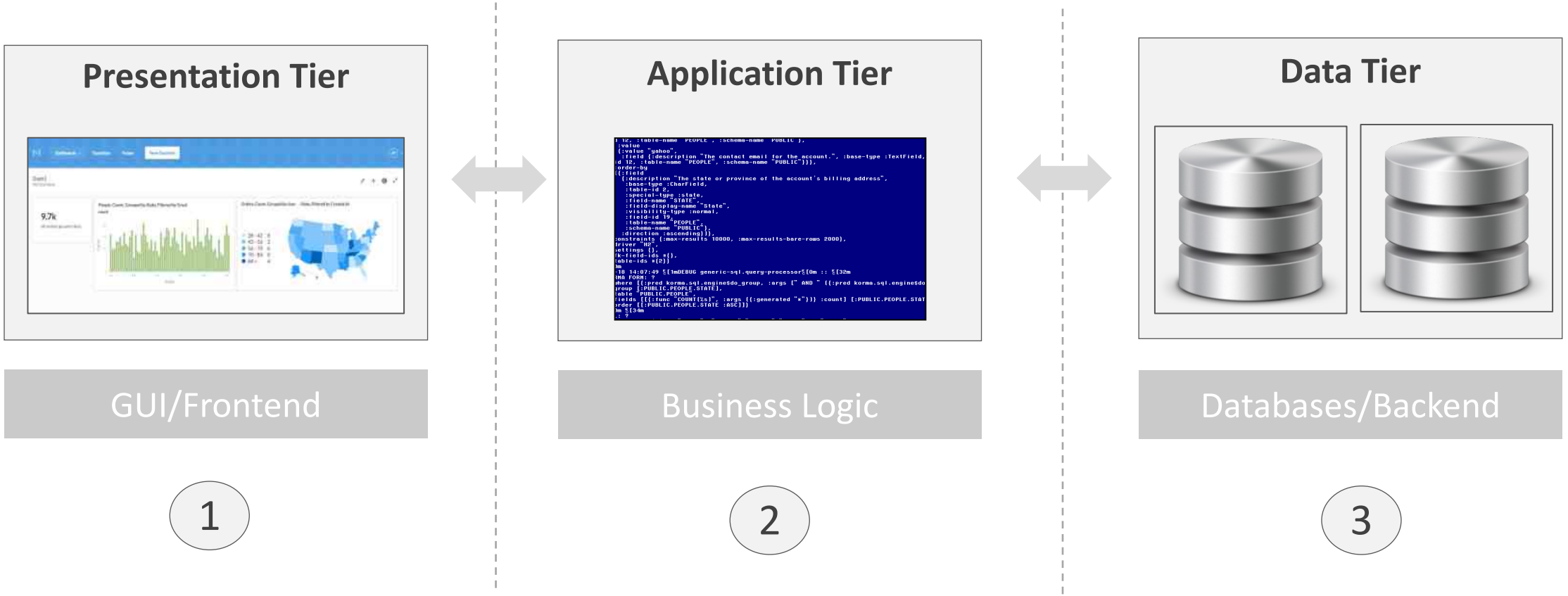
Bottom line: *Choose the right service for the workload, with security, reliability, and cost in mind.*



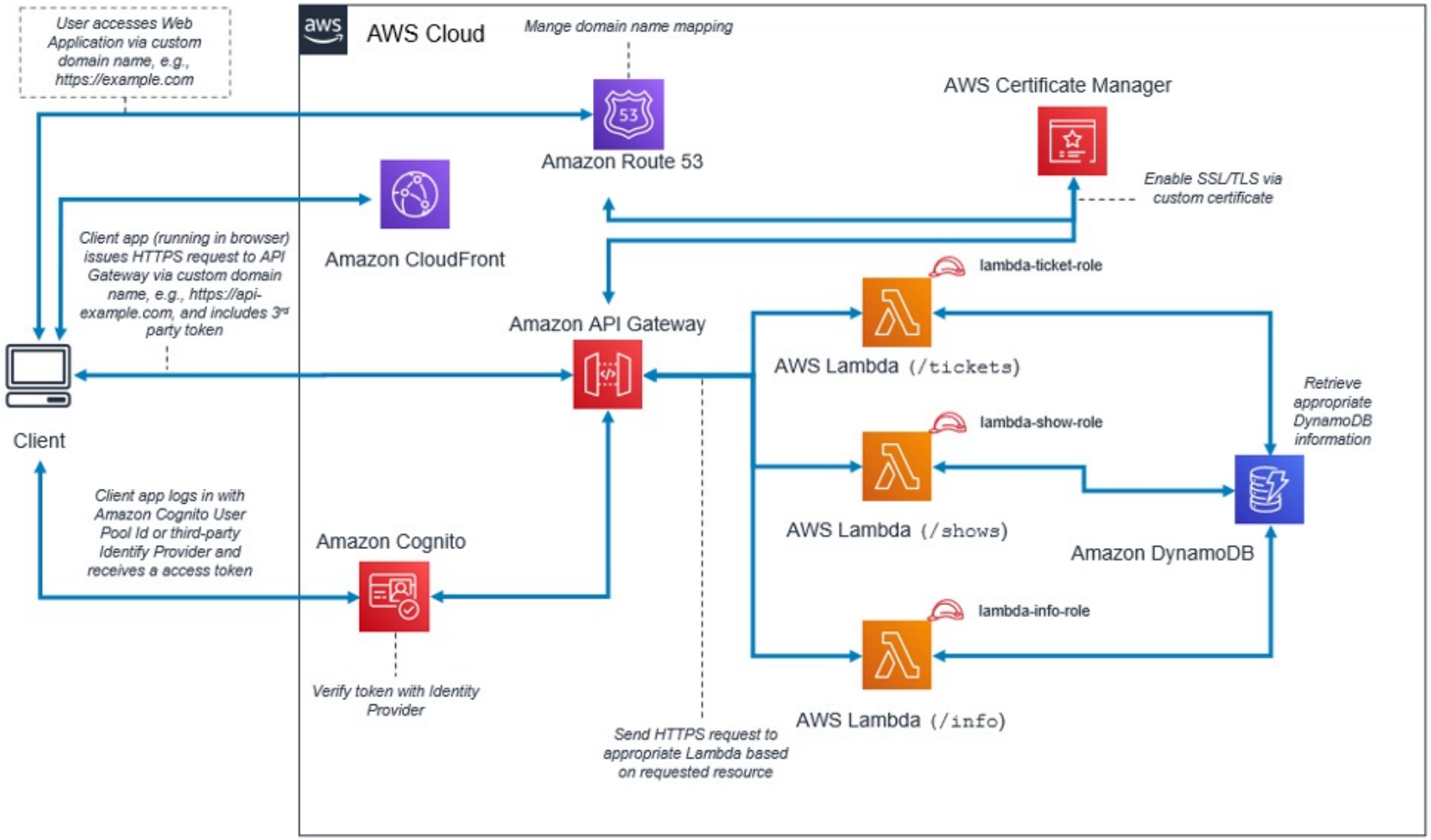
Reference Architectures

Module 7

3-tier Application Architecture

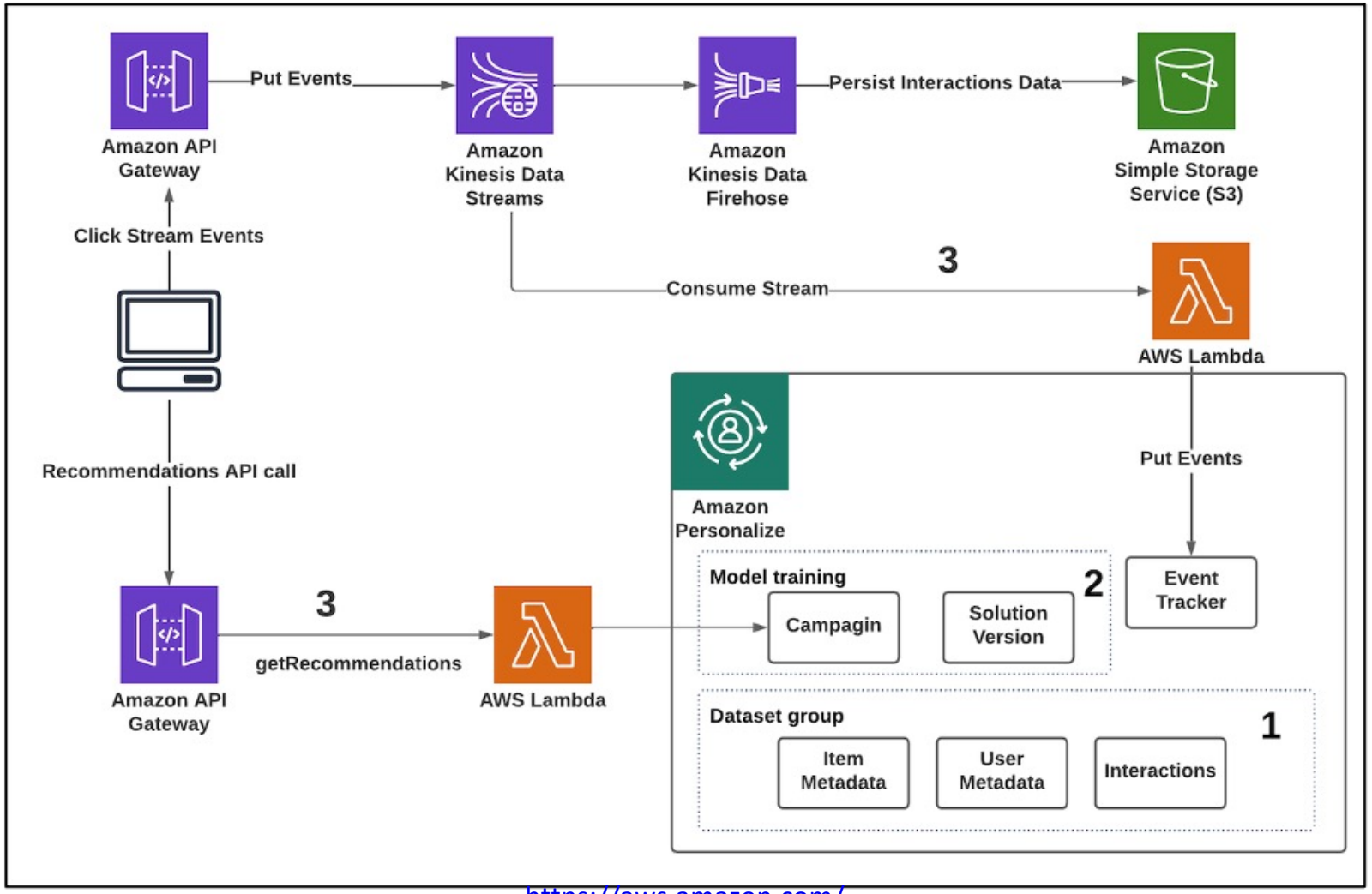


Web Application



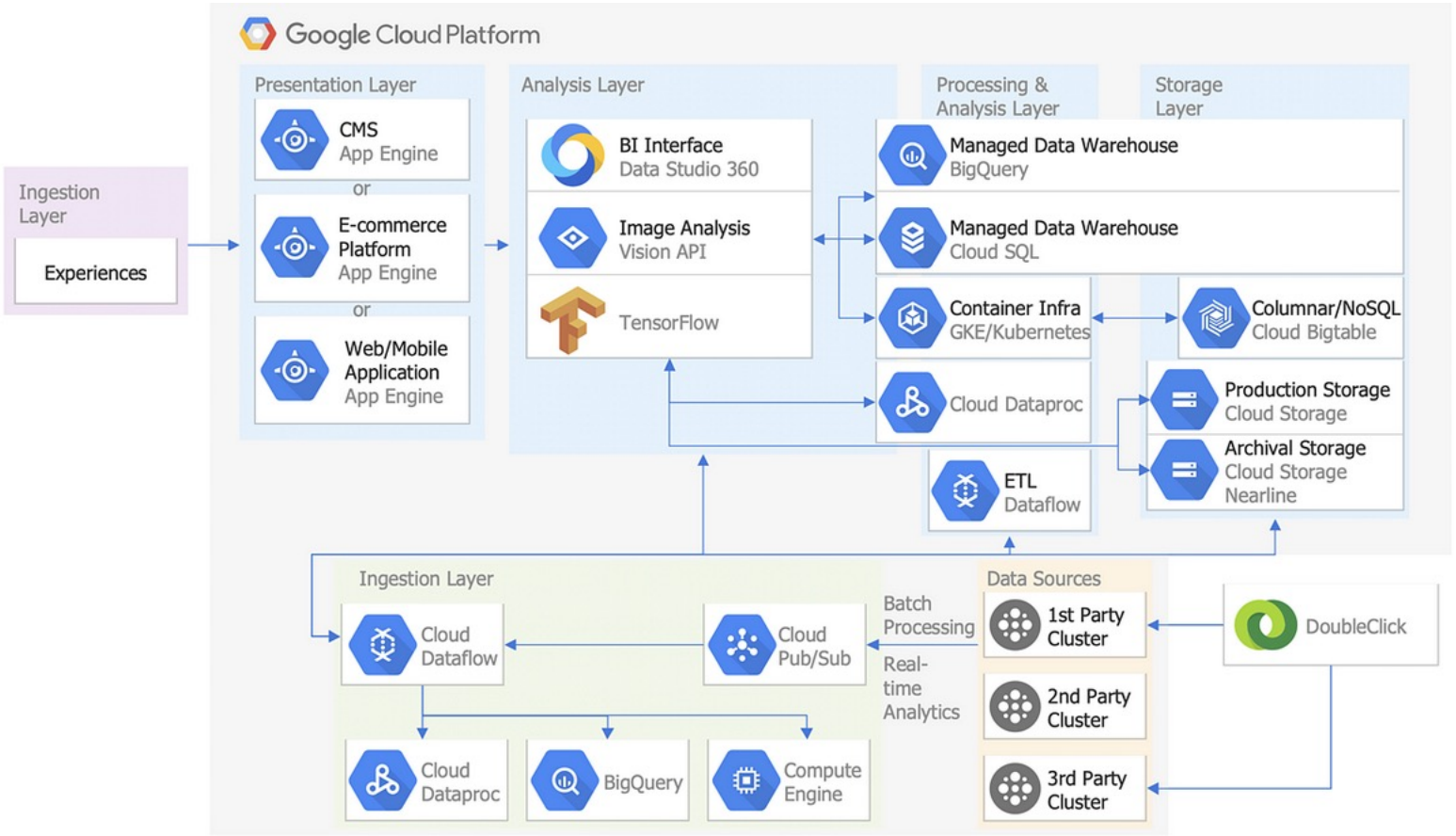
https://docs.aws.amazon.com

Recommendation System

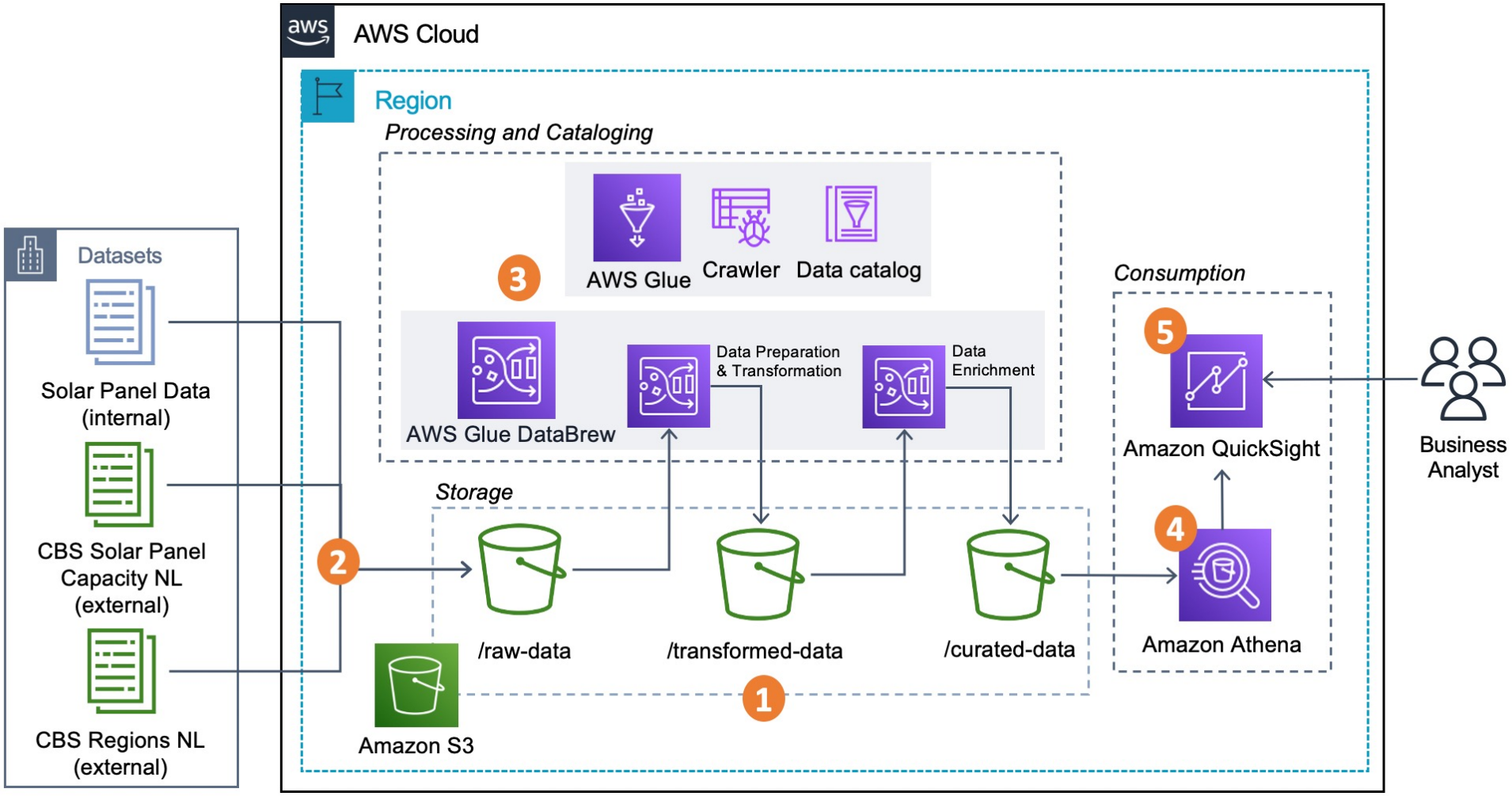


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Digital Marketing

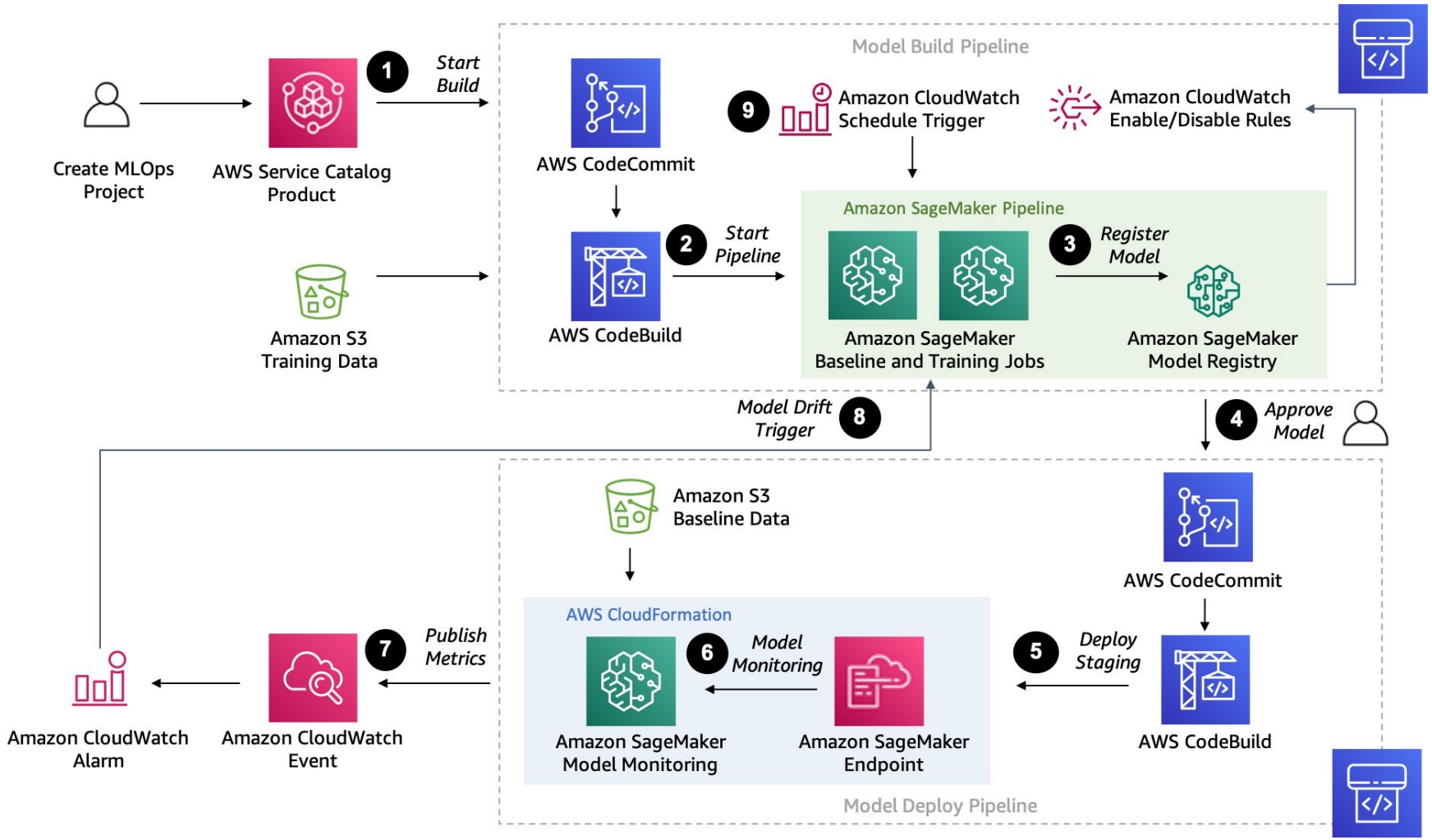


Data Analytics Pipeline



https://aws.amazon.com

ML Pipeline



https://aws.amazon.com



System Design

Module 8

Goal

- Build a scalable, low-latency video recommendation system on AWS that serves 20 personalized recommendations per user in under 150 ms, learns continuously from user behavior, and operates at a cost below $0.0005 per recommendation.
- Example:
  - YouTube: 2 billion logged-in users/month
  - 800M+ videos

Planning and Design

- Decompose a large-scale recommendation system into independently deployable AWS components
- Choose the right AWS storage service for each access pattern (OLTP, OLAP, cache, feature store, event stream)
- Design a two-stage ML recommendation pipeline (candidate generation → ranking)
- Quantitatively estimate and optimize monthly AWS cloud costs for a 10 M MAU service
- Articulate concrete design trade-offs (consistency vs. availability, cost vs. latency, simplicity vs. accuracy)
- Apply architectural best practices to making decisions

Design Steps

1. Requirements
2. High-Level Architecture
3. Data & Training Pipeline Design
4. API, Caching
5. Cost Estimates
6. Trade-offs, Resilience & Scale

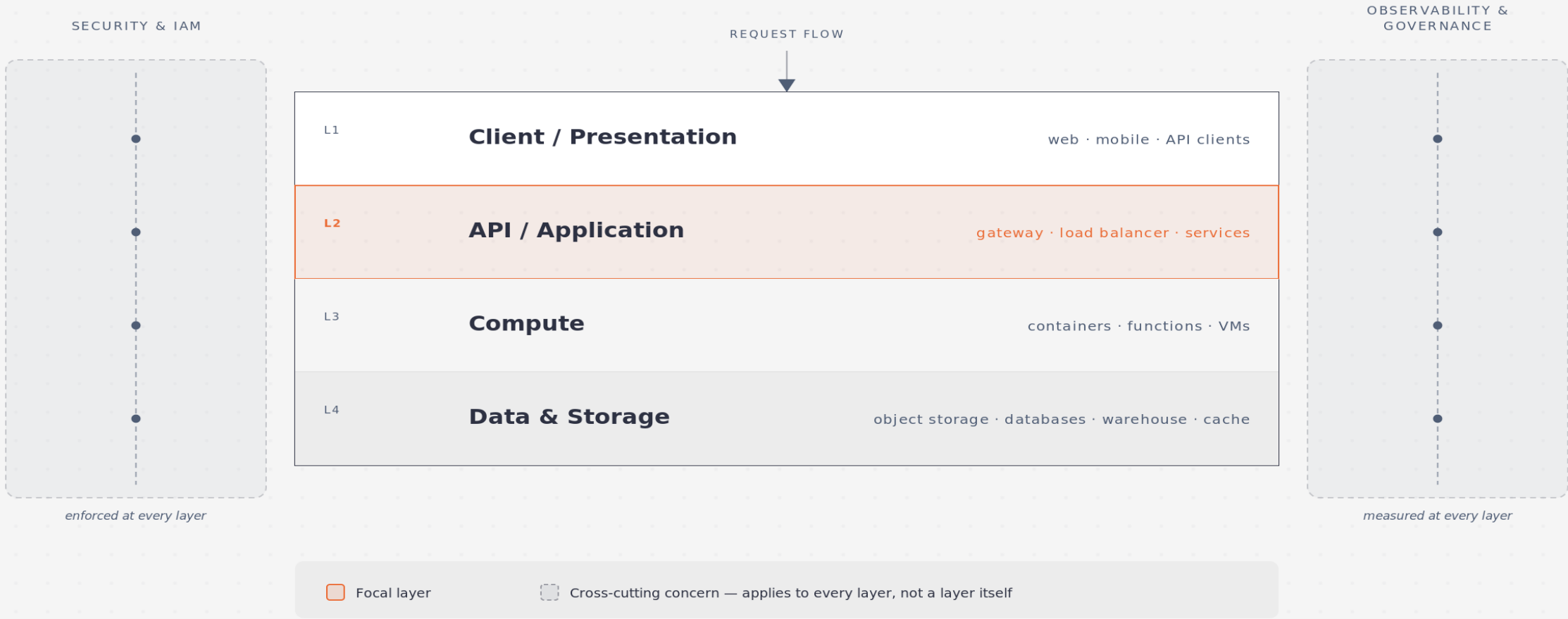
Functional Requirements

- Return N personalized recommendations (N = 20)
- Update recommendations near-real-time after each event
- Cold-start: new users and new items
- Filter watched, disliked, inappropriate content
- Contextual signals: time-of-day, device, location
- Trending / category-based fallback feeds
- REST API consumed by web, iOS, Android
- Log all recommendations for offline analysis / A/B testing

Non Functional Requirements

- Latency p99 < 150 ms
- Availability 99.99%
- Throughput 10K RPS peak
- Model staleness < 1 hour
- Feature freshness < 5 min for watch events
- Cost efficiency < $0.0005 per recommendation
- Data privacy PII anonymized (GDPR)
- Scalability Auto-scale to 10× in < 5 min

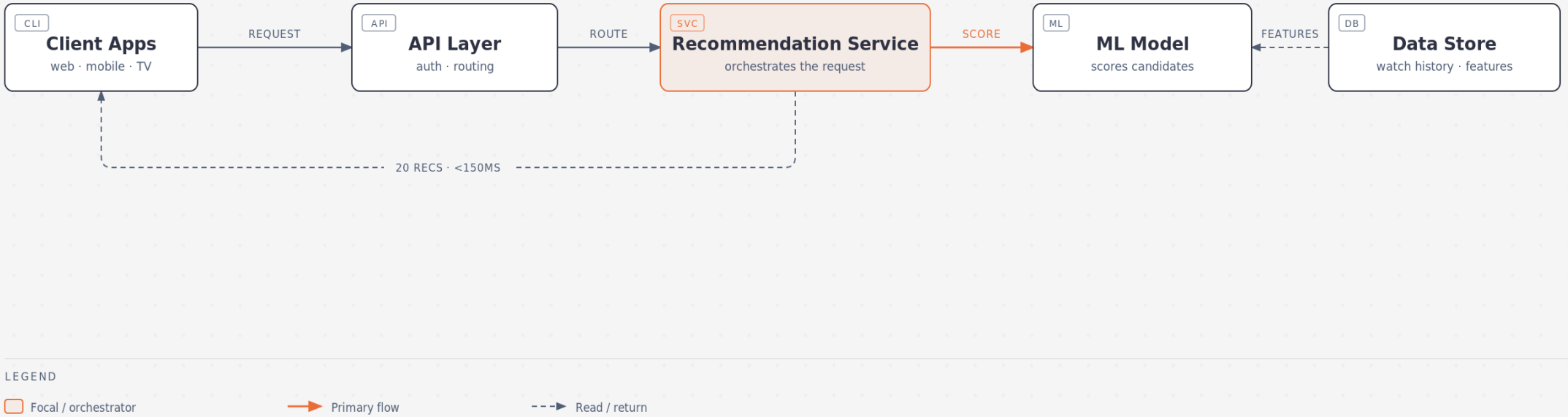
Modern Cloud Architecture — Layered View



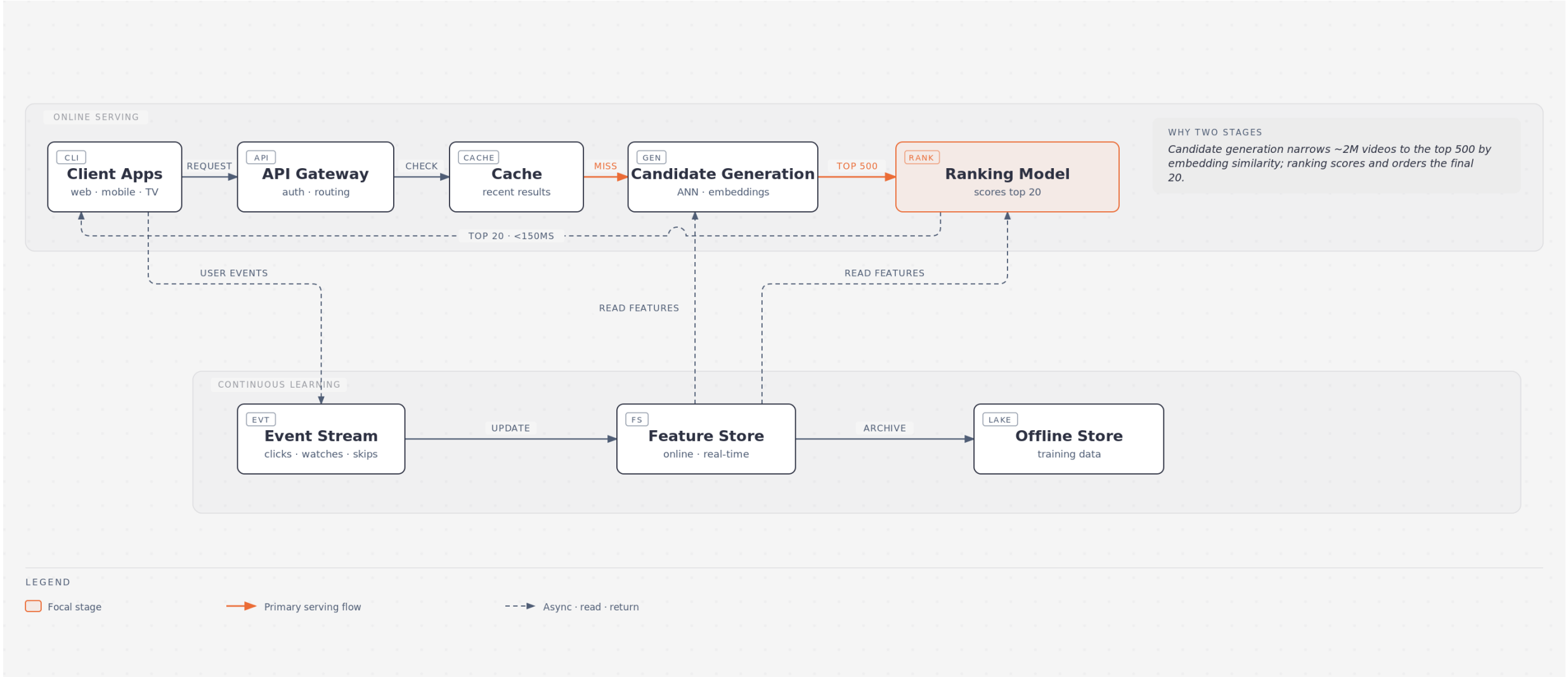
Request Path

NOTE

This view shows the forward request path only. The feedback loop that lets the system learn continuously from user behavior is added in the next view.



Two-Stage ML Pipeline



AWS Deployment, Latency & Cost Budget

